C Programming Assignment – Question 4

4. Dollars and Cents

Question

Help add two amounts of US money. One dollar equals 100 cents. The input contains the dollars and cents of the first amount, followed by the dollars and cents of the second amount.

Sample Input

```
30
10
140
99
```

Sample Output

```
171
9
```

C Program

```c
#include <stdio.h>

int main(void) {
    int dollars1, cents1, dollars2, cents2;
    int totalCents, dollars, cents;

    printf("Enter first dollars and cents: ");
    scanf("%d%d", &dollars1, &cents1);

    printf("Enter second dollars and cents: ");
    scanf("%d%d", &dollars2, &cents2);

    totalCents = dollars1 * 100 + cents1
               + dollars2 * 100 + cents2;

    dollars = totalCents / 100;
    cents = totalCents % 100;

    printf("%d\n%d\n", dollars, cents);
    return 0;
}
```

Explanation

1. Convert both dollar amounts into cents.

2. Add all cents together.

3. Divide the total by 100 to get dollars.

4. Use the remainder after division by 100 to get cents.

5. Print dollars and cents on separate lines.

Note

The result is 171 dollars and 9 cents.